CardioCog: block diagram of the whole project

3-year prediction of dementia and cognitive decline from demographic, cardiovascular, cognitive, clinical, genetic and PET information (NACC)

1 Data sources

NACC clinical table
217,598 visits · 57,038 people
46 centres · 2005–2026

PET summary tables
amyloid Centiloids (~5,100 people)
tau SUVR (2,724 people)

Raw PET images
599 DICOM scans · 446 people
~100 GB (amyloid, tau, FDG)

2 Cleaning & harmonisation

Missing-value codes
cleaned column by column
114 variables mapped by hand

Visit filter
in-person visits
UDS form versions 1–3

Cognitive z-scores
norms from healthy training people
(age, sex, education) → 6 domains

No future information
fixed traits carried forward only
PET used only if ≤ 365 days before

3 48 features in 5 groups

- **8** Demographics
- **25** Cardiovascular & medical
- **9** Cognitive tests
- **3** Clinical stage
- **3** Genetics (APOE)
- **+4** PET (Tier 2 only) — Centiloids · amyloid status · 2 tau

8 + 25 + 9 + 3 + 3 = 48 model inputs

4 Outcomes, cohort & split

Dementia within 3 years
main outcome · 19,470 people
2,818 events (14.5%)

CDR-SB rise ≥ 1 point
decline outcome · 20,355 people
5,035 events

3-year change in CDR-SB
regression target
(visit within ±6 months of 3 years)

Participant-level split
train 70 · validation 15 · test 15
+ 9 of 46 centres held out (seed 2026)

5 Models

Tier 1 classifiers (48 features)
Logistic regression · Random forest
XGBoost ★ main · MLP · FT-Transformer

Modality-encoder fusion network
5 encoders → 32-d tokens → attention
→ risk head + CDR-SB change head

Modality dropout
each group hidden with p = 0.35
→ robust when data are missing

Longitudinal models
landmark visit + trajectory features
GRU over up to 6 visits

Tier 2: PET late fusion
regularised logistic regression on
clinical risk + PET measures

6 Explain & evaluate

TreeSHAP explanations
contribution of each feature
and each group to every estimate

Ablation
each group alone, added in order,
and removed from the full model

Metrics
AUROC · AUPRC · Brier · calibration
net benefit · 1,000-bootstrap 95% CI

Robustness
subgroups · Cox survival model
new seeds · sensitivity analyses

7 PET image pipeline (separate experiment)

DICOM → static
frames in tracer
window averaged

→ **Registration**
rigid + affine to MNI152
mutual information

→ **Amyloid SUVR**
cortex ÷ cerebellum
r = 0.98 vs PET core

→ **Amyloid status**
SUVR 0.982 · CNN 0.960
(292 scans, 5-fold CV)

→ **Grad-CAM**
maps diffuse →
not used as evidence

8 Key results

Dementia AUROC 0.94
0.95 at held-out centres · 0.84 within MCI

Decline AUROC 0.86
0.89 at held-out centres · all 5 models alike

Cardio adds ≈ 0.001 AUROC
earlier visits ≤ 0.005 · PET no clear gain

9 Clinician prototype

React + FastAPI (Streamlit fallback)
risk % vs similar people · SHAP reasons
plausibility checks · plain-language report

evidence quotes ↔

10 Evidence assistant (RAG)

11 papers → 447 passages
MiniLM embeddings + BM25 → rank fusion
→ Qwen2.5-1.5B with quoted citations

11 Independent verification (scripts/verify_results.py): 17 of 17 checks pass

- raw files unchanged
- no person or centre in two partitions
- features unchanged when later visits are deleted
- labels recomputed: 0 disagreements
- shuffled labels → AUROC 0.508
- reported numbers = saved predictions
- 2 leaks found & fixed (results moved in 3rd decimal)

Research prototype · not for clinical use · all numbers are aggregates